RAG Learning

Edexcel A-Level Physics — Core Practicals

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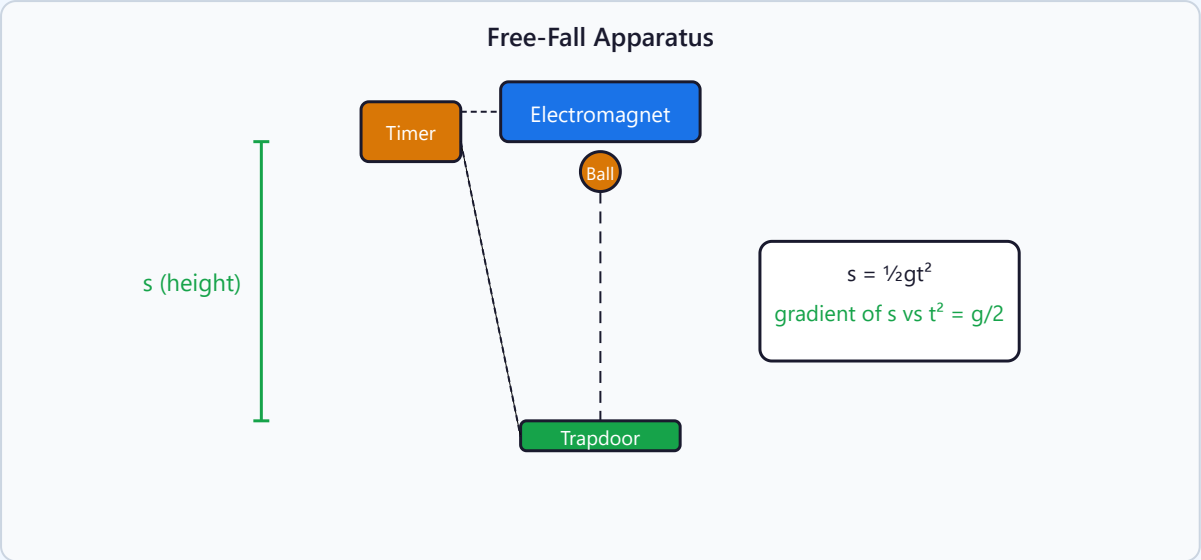
Note: CP15 (capacitor charge/discharge) is combined with CP7 as they cover the same practical.

⚗ CP1: Acceleration of a Freely Falling Object



Apparatus

- Electromagnet with release switch
- Steel ball bearing
- Trapdoor switch and electronic timer (or light gates)
- Metre ruler / 2 m ruler
- Clamp stand and boss

Method

1. Clamp the electromagnet at height s above the trapdoor.
2. When the switch is released, the timer starts; the ball falls and hits the trapdoor, stopping the timer.
3. Record time t. Repeat 3 times at each height and take the mean.
4. Vary s (at least 6 values, 0.2–1.0 m).
5. Plot s vs t^2. Gradient = g/2.

Safety

- Soft pad beneath trapdoor to catch ball
- Stable clamp stand; keep feet clear

Analysis

$$s = \frac{1}{2}gt^2 \Rightarrow \text{gradient of s vs } t^2 = g/2$$

Expected: $g \approx 9.81\ m\ s^{-2}$.

Source of Error	Type	Improvement
Residual magnetism delays ball release	Systematic	Use a non-magnetic release mechanism or demagnetiser
Air resistance on ball	Systematic	Use a small, dense ball
Height measurement uncertainty	Random	Use a 2 m ruler; measure from bottom of ball to trapdoor
Timer precision	Random	Use electronic timer; repeat and average

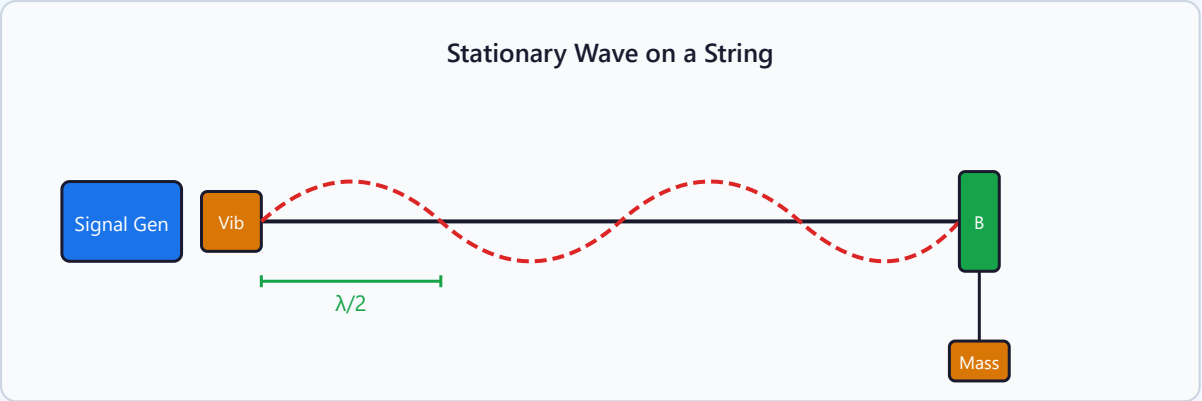
Exam Tip: A non-zero y-intercept on the s vs t^2 graph suggests systematic error (e.g. timing delay from residual magnetism). Light gates can remove this source of error entirely.

Worked Example: A student records the following data: s = 0.80 m, mean t = 0.404 s. Calculate g.

$s = \frac{1}{2}gt^2$, so $g = 2s/t^2 = 2 \times 0.80 / (0.404)^2 = 1.60 / 0.163 =$ **$9.8\ m\ s^{-2}$**

The percentage uncertainty in t is typically ~1-2% (from reaction time when using electronic timer), giving ~2-4% uncertainty in g (since t is squared).

⚗ CP2: Frequency of Stationary Waves on a String



Apparatus

- Signal generator and vibrator
- String (nylon thread), pulley, bench clamp
- Slotted masses and hanger (for tension)
- Metre ruler

Method

1. Attach the string to the vibrator, pass it over a pulley, and hang masses to provide tension T = mg.
2. Adjust the frequency until the fundamental mode (1 loop) is observed.
3. Record the frequency and measure the vibrating length L.
4. Increase frequency to find 2nd, 3rd, etc. harmonics.
5. Vary tension by changing masses and repeat.

Safety

- Safety screen in case string snaps
- Soft pad beneath hanging masses

Analysis

$$f = (1/2L)\sqrt{(T/\mu)}$$

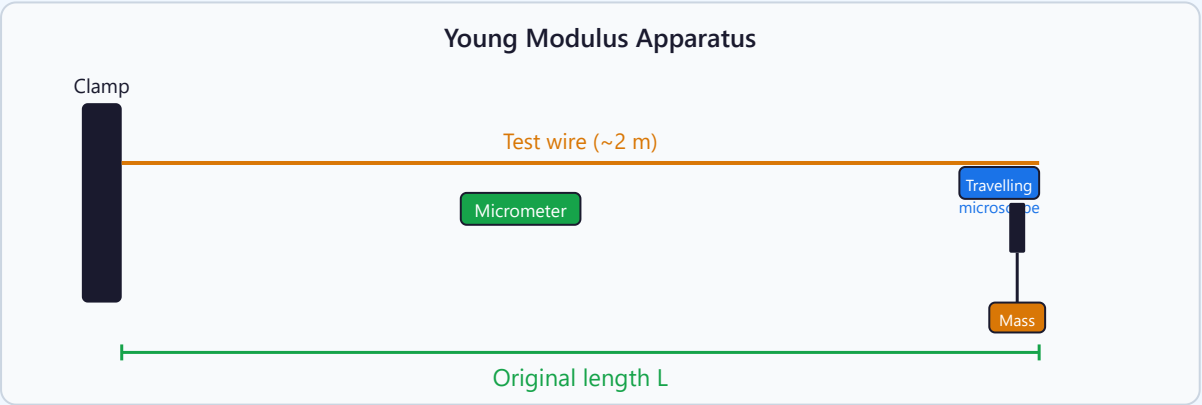
Plot f vs 1/L (constant T): straight line through origin, gradient = $(1/2)\sqrt{(T/\mu)}$.

Plot f vs $\sqrt{T}$ (constant L): straight line through origin.

Source of Error	Type	Improvement
Judging resonance by eye	Random	Use a paper rider; resonance flicks it off
String mass not negligible	Systematic	Include string mass when calculating μ
Measuring vibrating length	Random	Use a mm ruler; measure node to node
Mass hanger mass omitted	Systematic	Include hanger mass in tension

Exam Tip: The vibrator is not quite at a node, so the measured L is slightly longer than the true resonant length. Measure between nodes rather than from the vibrator end for better accuracy.

⚗ CP3: Young Modulus of a Wire



Apparatus

- Long thin wire (copper, ~2 m)
- Micrometer screw gauge
- Travelling microscope or vernier scale (for extension)
- Slotted masses and hanger
- Tape measure, G-clamp, safety goggles

Method

1. Measure the wire diameter at several points using a micrometer. Calculate mean d and $A = \pi d^2/4$.
2. Measure original length L with tape measure.
3. Add masses in steps (e.g. 100 g). After each addition, measure extension with travelling microscope.
4. Record $F = mg$ and ΔL at each step.
5. Unload to check elastic behaviour (wire returns to original length).
6. Plot stress vs strain. Gradient = Young modulus E.

Safety

- Wear safety goggles — wire may snap
- Cushion or sand tray beneath masses
- Do not exceed the elastic limit

Analysis

$$E = \text{stress} / \text{strain} = (F/A) / (\Delta L/L) = FL / (A\Delta L)$$

Plot stress (σ) vs strain (ε). Gradient of linear region = E.

Source of Error	Type	Improvement
Wire diameter varies	Random	Multiple readings at different positions; use mean
Kinks in wire	Systematic	Apply small initial load to straighten wire
Thermal expansion	Systematic	Use a reference wire for compensation
Small extensions hard to measure	Random	Use travelling microscope; use long, thin wire

Exam Tip: The percentage uncertainty in diameter is the dominant source because d appears squared in the area formula. Always use a micrometer and take at least 6 readings at

different positions and orientations.

Worked Example: A copper wire has: L = 2.50 m, d = 0.28 mm, extension = 1.8 mm under a load of 30 N. Calculate the Young modulus.

$A = \pi d^2/4 = \pi(0.28 \times 10^{-3})^2/4 = 6.16 \times 10^{-8}\ m^2$

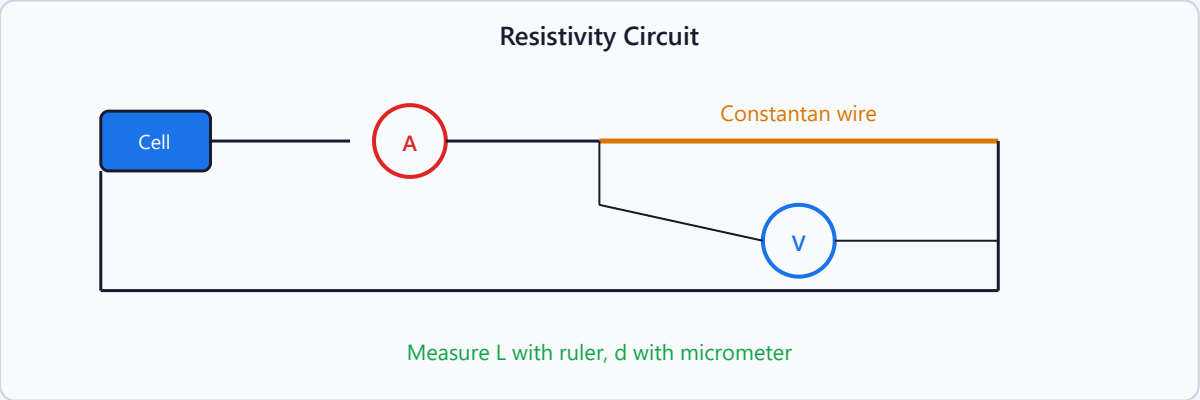
Stress = F/A = $30 / 6.16 \times 10^{-8} = 4.87 \times 10^{8}$ Pa

Strain = $\Delta L/L = 1.8 \times 10^{-3} / 2.50 = 7.2 \times 10^{-4}$

E = stress/strain = $4.87 \times 10^{8} / 7.2 \times 10^{-4}$ = **6.8×10^{11} Pa**

(Accepted value for copper: ~1.1×10^{11} Pa — the discrepancy suggests the diameter was measured inaccurately, since it has the largest percentage uncertainty.)

⚗ CP4: Resistivity of a Wire



Apparatus

- Constantan wire (~1 m, taped to ruler)
- Ammeter and voltmeter
- Cell/power supply and switch
- Micrometer screw gauge
- Metre ruler, crocodile clips

Method

1. Measure wire diameter at several points using micrometer. Calculate $A = \pi d^2/4$.
2. Set up circuit: cell, switch, ammeter in series with wire; voltmeter across wire.
3. Use clips to select length L. Record V and I. Calculate $R = V/I$.
4. Repeat for 6+ lengths (0.1–1.0 m).
5. Plot R vs L. Gradient = ρ/A, so ρ = gradient × A.

Safety

- Switch off between readings to prevent heating
- Low voltage (< 6 V); do not touch energised wire

Analysis

$$R = \rho L/A \Rightarrow \text{plot R vs L, gradient} = \rho/A$$

Expected: ρ (constantan) ≈ 4.9×10^{-7} Ω m.

Source of Error	Type	Improvement
Wire heating changes R	Systematic	Switch off between readings; use low current
Contact resistance at clips	Systematic	Clean contacts; ensure firm grip
Diameter variation	Random	Multiple readings; take mean
Length measurement at clips	Random	Tape wire to ruler; align clips carefully

Exam Tip: Constantan is chosen because its resistivity barely changes with temperature, unlike copper. The graph of R vs L should pass through the origin; a non-zero intercept suggests contact resistance.

⚗ CP5: EMF and Internal Resistance

EMF and Internal Resistance

$V = \varepsilon - Ir \Rightarrow$ y-intercept = ε, gradient = −r

Apparatus

- Cell (1.5 V), variable resistor, ammeter, voltmeter
- Switch, connecting wires

Method

1. Connect cell, ammeter, variable resistor in series. Voltmeter across cell terminals.
2. For each resistance setting, record V and I.
3. Repeat for 6+ settings. Switch off between readings.
4. Plot V vs I.

Safety

- Do not short-circuit the cell
- Switch off between readings to preserve cell

Analysis

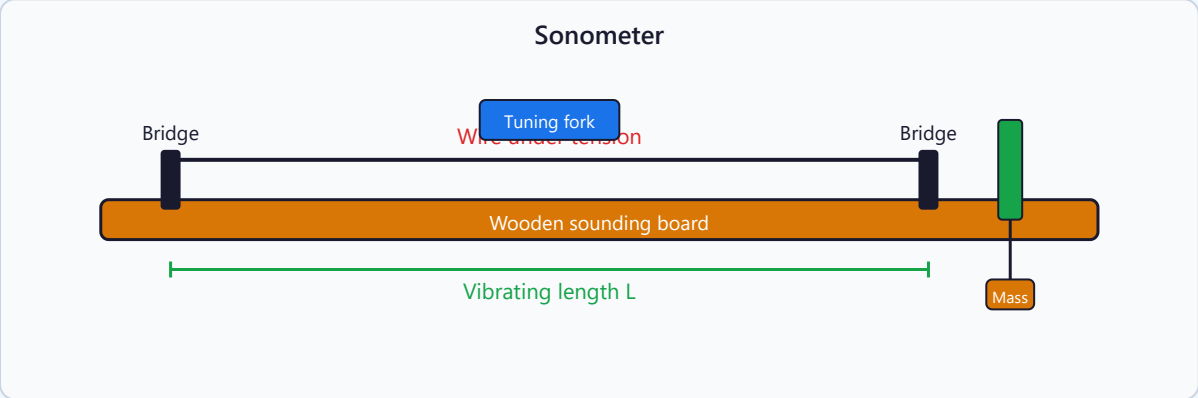
$V = \varepsilon - Ir$. Plot V vs I: y-intercept = ε, gradient = −r.

Source of Error	Type	Improvement
Cell runs down	Systematic	Switch off between readings; work quickly
Ammeter resistance	Systematic	Use low-resistance digital ammeter
Voltmeter draws current	Systematic	Use high-impedance digital voltmeter

Fluctuating readings	Random	Wait for stable readings; take repeats

Exam Tip: The x-intercept gives the short-circuit current ε/r. The gradient magnitude gives internal resistance. Typical r for a cell: 0.5–2 Ω.

⚗ CP6: Sonometer Investigation



Apparatus

- Sonometer (wooden board with wire and movable bridges)
- Set of tuning forks (known frequencies)
- Slotted masses and hanger (tension)
- Metre ruler, paper rider

Method

1. Set a known tension T by hanging masses from one end of the wire.
2. Strike a tuning fork and hold it near the wire. Move one bridge until the wire resonates (paper rider jumps off).
3. Measure the resonant length L between the bridges.
4. Repeat with different tuning forks (different frequencies) or different tensions.
5. Plot f vs 1/L (constant T) or f vs √T (constant L).

Safety

- Safety glasses in case wire snaps
- Do not over-tension the wire
- Secure masses so they cannot fall

Analysis

$$f = (1/2L)\sqrt{(T/\mu)}$$

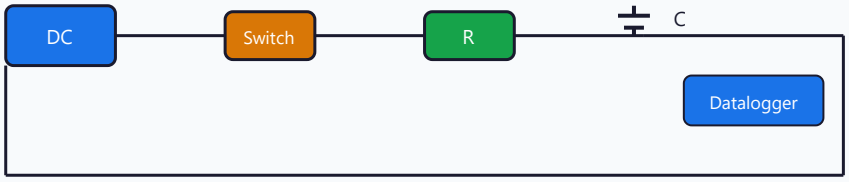
Plot f vs 1/L: straight line through origin. Gradient = (1/2)√(T/μ).

Source of Error	Type	Improvement
Difficulty judging exact resonance	Random	Use a paper rider to detect resonance clearly
Bridge position measurement	Random	Use mm ruler; align carefully
Wire may not vibrate in fundamental	Systematic	Pluck gently at centre; listen for pure tone
Tuning fork frequency uncertainty	Systematic	Use calibrated, high-quality forks

Exam Tip: A sonometer differs from the vibrator method (CP2) in that resonance is matched to external tuning forks rather than driven by a signal generator. Both investigate the same equation f = (1/2L)√(T/μ).

⚗ CP7/CP15: Capacitor Charge and Discharge

RC Discharge Circuit



$V = V_0e^{(-t/RC)} \Rightarrow \ln(V)$ vs t: gradient = $-1/RC$

Apparatus

- Capacitor (e.g. 470–2200 μF electrolytic)
- Resistor (chosen so $\tau = RC$ is 10–30 s)
- DC power supply, two-way switch
- Voltmeter or datalogger, stopwatch

Method

1. Connect C and R in series with supply and two-way switch. Datalogger across C.
2. Charge: flip switch to supply side. Record V vs t as capacitor charges.
3. Discharge: flip switch to disconnect supply. Record V vs t during discharge.
4. Plot ln(V) vs t for discharge. Gradient = $-1/RC$.

Safety

- Correct polarity for electrolytic capacitors
- Discharge capacitor before handling
- Low voltage (< 12 V)

Analysis

$$V = V_0e^{-t/RC} \Rightarrow \ln V = \ln V_0 - t/(RC)$$

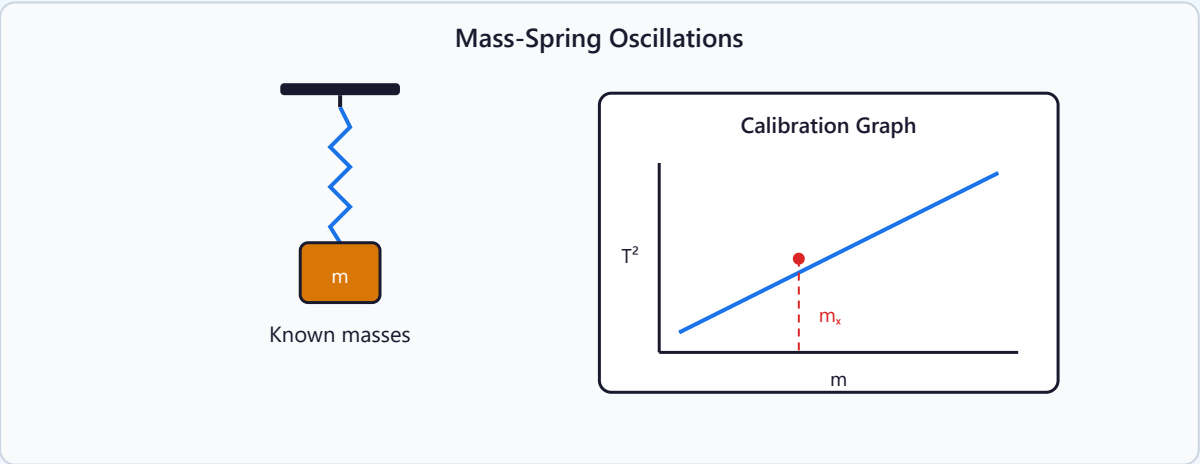
Plot ln(V) vs t: straight line, gradient = $-1/RC$, y-intercept = $\ln V_0$.

Time constant $\tau = RC$ is time for V to fall to 37% of V_0.

Source of Error	Type	Improvement
Voltmeter draws current	Systematic	Use high-impedance voltmeter or datalogger
Manual timing errors	Random	Use datalogger for continuous recording
Capacitor leakage	Systematic	Use high-quality capacitor
Temperature changes	Systematic	Conduct in stable temperature environment

Exam Tip: Choose R and C so the time constant $\tau = RC$ is convenient (10–30 s). After 5τ, the capacitor is effectively fully charged or discharged. The charge on the capacitor is $Q = CV$, so you can also plot $\ln(Q)$ vs t.

CP8: Unknown Mass from Spring Oscillation Period



Apparatus

- Helical spring, clamp stand
- Set of known slotted masses
- Unknown mass
- Stopwatch

- Ruler (optional)

Method

1. Hang a known mass from the spring. Pull down slightly and release.
2. Time 10 oscillations. Calculate T. Calculate T^2.
3. Repeat for 6+ known masses.
4. Plot T^2 vs m (calibration graph). Should be a straight line through the origin.
5. Hang the unknown mass, measure T, calculate T^2.
6. Read off the unknown mass from the calibration graph.

Safety

- Secure clamp stand with G-clamp
- Do not overload the spring beyond its elastic limit
- Keep clear of oscillating masses

Analysis

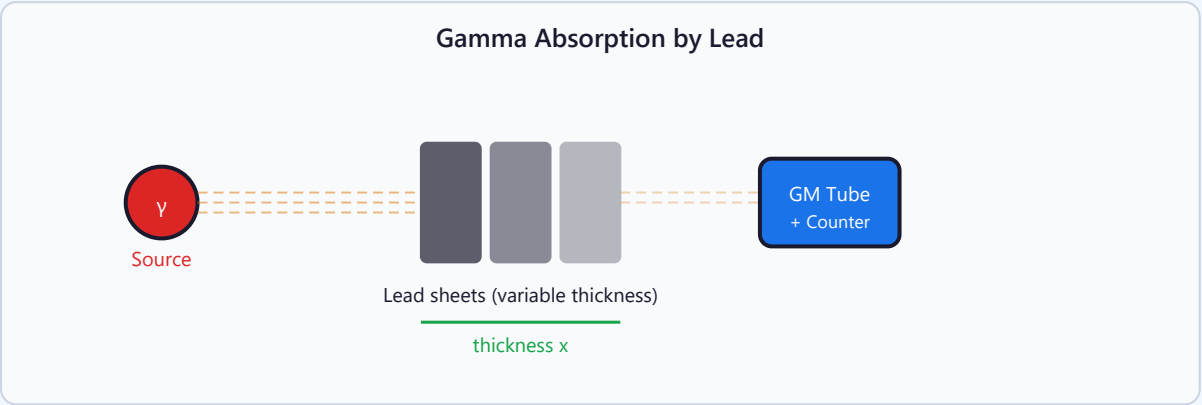
$$T = 2\pi\sqrt{(m/k)} \Rightarrow T^2 = (4\pi^2/k)m$$

Gradient of T^2 vs m = $4\pi^2/k$. Read unknown mass from the graph using its T^2 value.

Source of Error	Type	Improvement
Miscounting oscillations	Random	Use fiducial marker; count from equilibrium
Non-vertical oscillation	Random	Release carefully from rest; use guide tube
Spring mass not negligible	Systematic	Add $m_{eff} = m_{spring}/3$ to the oscillating mass
Reaction time	Random	Time 10+ oscillations

Exam Tip: If the graph does not pass through the origin, the spring has an effective mass of its own (approximately 1/3 of the spring mass). The x-intercept gives this effective mass as a negative offset.

⚗ CP9: Absorption of Gamma Radiation by Lead



Apparatus

- Gamma source (e.g. cobalt-60, sealed)
- GM tube and counter/ratemeter
- Set of lead sheets (varying thickness)
- Ruler or vernier callipers (to measure lead thickness)
- Source holder and clamp

Method

1. Measure background count rate (no source, count for 60+ seconds).
2. Place source at fixed distance from GM tube. Record count rate with no absorber.
3. Insert lead sheets of known thickness between source and detector.
4. Record corrected count rate (subtract background) for each thickness.
5. Repeat for 6+ thicknesses.
6. Plot ln(corrected count rate) vs thickness x.

Safety

- Handle source with tongs; never bare hands
- Keep maximum distance; minimise exposure time
- Store source in lead-lined container
- Follow ALARA principle

Analysis

Intensity follows: $I = I_0 e^{-\mu x}$

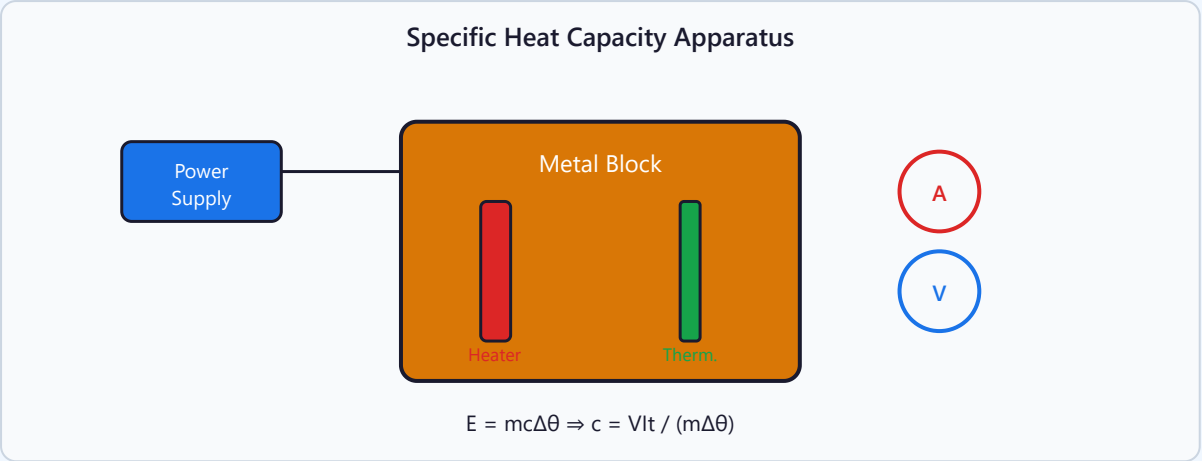
Plot **ln(C) vs x**: straight line with gradient = −μ (linear attenuation coefficient).

Half-value thickness: $x_{1/2} = \ln 2 / \mu$.

Source of Error	Type	Improvement
Random nature of decay	Random	Count for long periods; take repeats
Background radiation fluctuates	Random	Multiple background readings; long count time
Lead sheets not uniform thickness	Systematic	Measure thickness at multiple points with callipers
Scattered radiation reaching detector	Systematic	Collimate the beam; use narrow source-detector geometry

Exam Tip: The half-value thickness is the absorber thickness needed to halve the intensity. For gamma radiation in lead, this is typically a few mm to a few cm depending on the energy of the gamma photons.

⚗ CP10: Specific Heat Capacity



Apparatus

- Metal block (aluminium or copper) with holes for heater and thermometer
- Immersion heater
- Thermometer (or temperature probe)
- Ammeter, voltmeter
- Power supply, stopwatch
- Balance (to measure block mass)
- Insulation (lagging) for the block

Method

1. Measure the mass of the metal block using a balance.
2. Insert the heater and thermometer into the block. Add a drop of oil to ensure good thermal contact.
3. Record the initial temperature θ_1.
4. Switch on the heater. Record V, I, and time t.
5. Heat for a set time (e.g. 10 minutes). Record final temperature θ_2.
6. Calculate: $c = VIt / (m\Delta\theta)$, where $\Delta\theta = \theta_2 - \theta_1$.

Safety

- Metal block gets very hot — use tongs or heat-resistant gloves
- Do not touch the heater element

- Ensure heater is fully inserted to avoid overheating exposed parts
- Place block on a heat-resistant mat

Analysis

Energy supplied: E = VIt. Energy gained: E = mcΔθ.

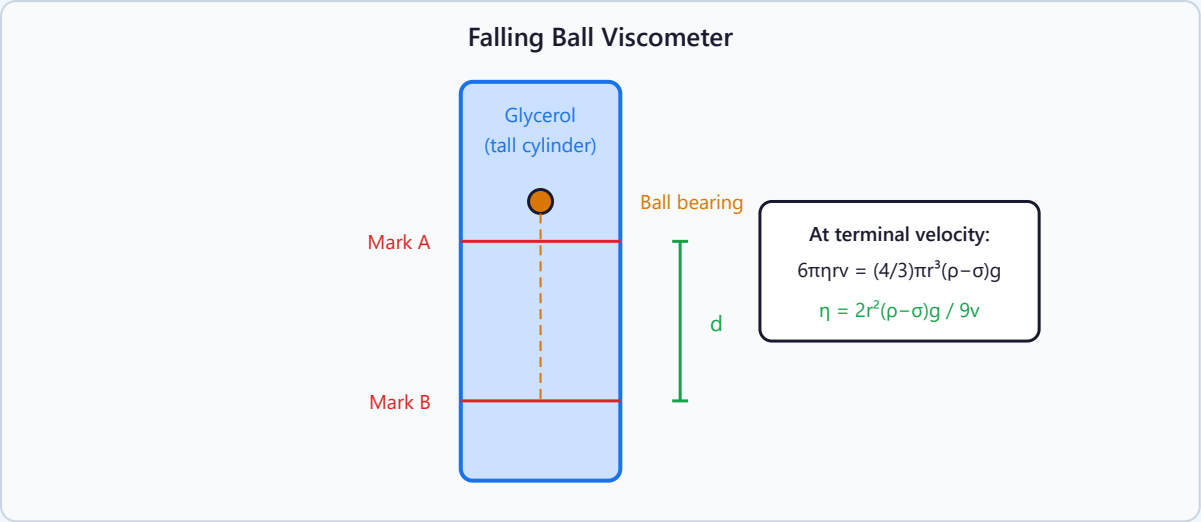
$$c = VIt / (m\Delta\theta)$$

Alternatively, plot Δθ vs t. Gradient = VI/(mc), so c = VI/(m × gradient).

Source of Error	Type	Improvement
Heat loss to surroundings	Systematic (c appears too high)	Insulate the block with lagging
Thermometer not at thermal equilibrium with block	Systematic	Use oil in thermometer hole; wait for steady reading
Energy lost heating heater itself	Systematic	Ensure heater is small relative to block
Voltage/current fluctuations	Random	Monitor V and I throughout; use regulated supply

Exam Tip: Heat losses mean the measured c is always slightly higher than the true value, because some electrical energy is lost to the surroundings rather than heating the block. Insulation and a cooling correction can reduce this.

⚗ CP11: Viscosity by Falling Ball (Stokes' Law)



Apparatus

- Tall measuring cylinder filled with glycerol (or viscous oil)
- Small steel ball bearings (various sizes)
- Micrometer (to measure ball diameter)
- Stopwatch
- Ruler (for distance between timing marks)
- Thermometer
- Elastic bands (for timing marks on the cylinder)

Method

1. Measure the diameter of the ball bearing using a micrometer. Calculate radius r.
2. Mark two reference lines on the cylinder (well below the surface so the ball reaches terminal velocity).
3. Gently drop the ball into the centre of the liquid surface.
4. Time how long the ball takes to travel between the two marks. Calculate terminal velocity v = d/t.
5. Repeat 3 times and take the mean.
6. Repeat with different ball sizes to investigate how η depends on r.
7. Record the temperature of the liquid (viscosity is temperature-dependent).

Safety

- Clean up spilled glycerol immediately (slipping hazard)
- Retrieve balls with a magnet to avoid tipping the cylinder
- Ensure the cylinder is stable and cannot topple

Analysis

At terminal velocity, drag force = weight − upthrust:

$$6\pi\eta rv = (4/3)\pi r^3(\rho - \sigma)g$$

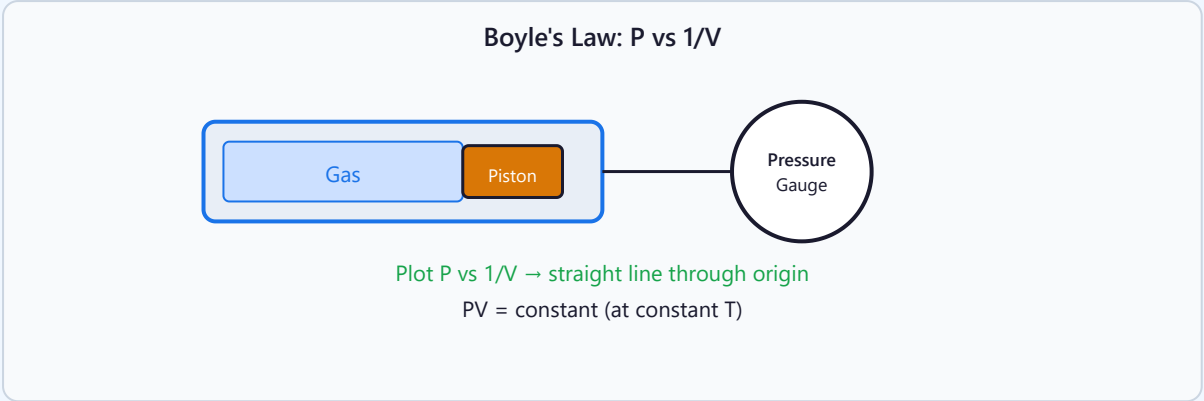
$$\eta = 2r^2(\rho - \sigma)g \,/\, (9v)$$

where ρ = density of ball, σ = density of fluid.

Source of Error	Type	Improvement
Ball not at terminal velocity between marks	Systematic	Place top mark well below the surface
Ball not falling centrally (wall effects)	Systematic	Use a wide cylinder; drop ball in the centre
Temperature changes affect viscosity	Systematic	Monitor temperature; work quickly
Timing reaction time	Random	Use a longer distance between marks; video analysis

Exam Tip: Stokes' law only applies at low Reynolds numbers (laminar flow). Use small, slow-moving balls in viscous fluid. The ball must reach terminal velocity before timing begins, so place the top marker well below the liquid surface.

⚗ CP12: Boyle's Law



Apparatus

- Sealed gas syringe or Boyle's law apparatus
- Bourdon pressure gauge
- Thermometer (to verify constant temperature)

Method

1. Record initial volume V and pressure P.
2. Slowly change the volume by pushing/pulling the piston. Wait for temperature to equilibrate.
3. Record new V and P. Repeat for 6+ data points.
4. Plot P vs 1/V. Expect a straight line through the origin.

Safety

- Do not over-pressurise
- Use safety screen with pressurised apparatus

Analysis

$$PV = nRT = \text{constant (at constant T)}$$

P vs 1/V: straight line through origin. Gradient = nRT.

Source of Error	Type	Improvement
Gas temperature changes on compression	Systematic	Compress slowly; wait for equilibrium

Friction in piston	Systematic	Lubricate; tap gently
Gas leaks	Systematic	Check seals with soap solution
Parallax on pressure gauge	Random	Read at eye level; use digital gauge

Exam Tip: Plotting P vs 1/V gives a straight line that is easier to verify than plotting PV vs P (which should be a horizontal line). Ensure temperature stays constant; even small temperature changes will affect results.

⚗ CP13: Thermistor Calibration in Potential Divider

$_s$ R_1 NTC V

Water bath V_o^{54} varies with temperature Plot V_o^{54} vs θ for calibration

Apparatus

- NTC thermistor
- Fixed resistor R_1
- DC power supply
- Voltmeter (across thermistor)
- Water bath, thermometer
- Beaker, stirrer, heat source

Method

1. Set up the potential divider: supply → R_1 → thermistor → back to supply. Voltmeter across the thermistor.
2. Place the thermistor in a water bath at a known temperature.
3. Record the temperature θ and voltage V across the thermistor.
4. Heat the water bath slowly. Record V at regular temperature intervals (e.g. every 5 °C).
5. Continue from room temperature to ~80 °C.
6. Plot V vs θ to create a calibration curve.

Safety

- Handle hot water carefully; do not overheat
- Electrical components must be kept dry
- Use a heat-resistant mat

Analysis

For an NTC thermistor, resistance decreases as temperature rises. In the potential divider:

$$V_{out} = V_s \times R_T / (R_1 + R_T)$$

As temperature rises, R_T falls, so V_{out} decreases. The calibration curve allows unknown temperatures to be determined from voltage readings.

Source of Error	Type	Improvement
Temperature not uniform in water bath	Systematic	Stir the water continuously
Thermistor response time lag	Systematic	Heat slowly; allow time for equilibrium
Self-heating of thermistor by current	Systematic	Use a small current (high R_1)
Thermometer reading uncertainty	Random	Use a digital thermometer with 0.1 °C resolution

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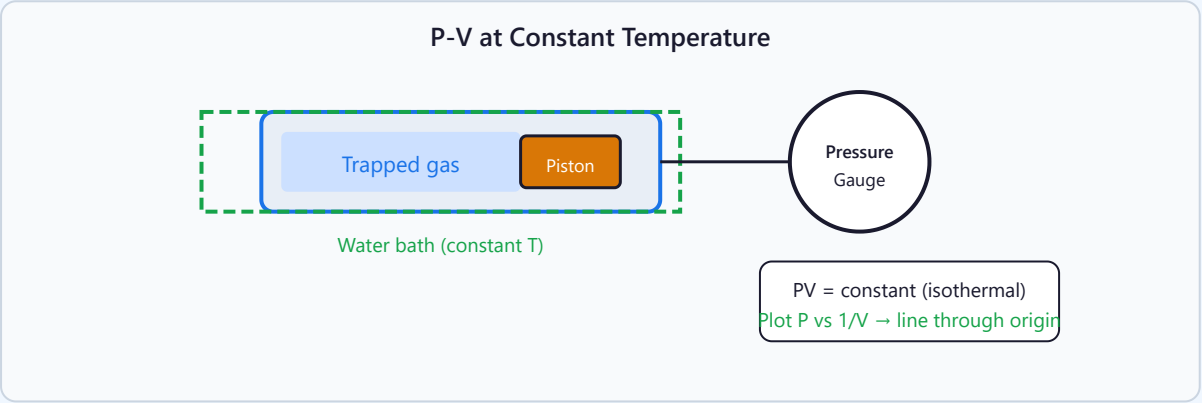
Exam Tip: The sensitivity of the thermistor circuit depends on the choice of R_1. Maximum sensitivity occurs when R_1 is similar in value to the thermistor resistance at the temperature of interest. NTC means "negative temperature coefficient."

CP14: Pressure-Volume at Fixed Temperature



Apparatus

- Gas syringe or Boyle's law apparatus immersed in water bath
- Pressure gauge
- Thermometer
- Constant temperature water bath (thermostatically controlled if possible)

Method

1. Immerse the gas apparatus in a constant-temperature water bath.
2. Allow sufficient time for thermal equilibrium.
3. Record V and P at this temperature.
4. Change V slowly. Wait for temperature to re-equilibrate. Record new P.
5. Repeat for 6+ volumes at the same temperature.
6. Optionally repeat at a different temperature for comparison.
7. Plot P vs 1/V.

Safety

- Do not over-pressurise apparatus
- Handle hot water carefully if using elevated temperatures
- Safety screen near pressurised equipment

Analysis

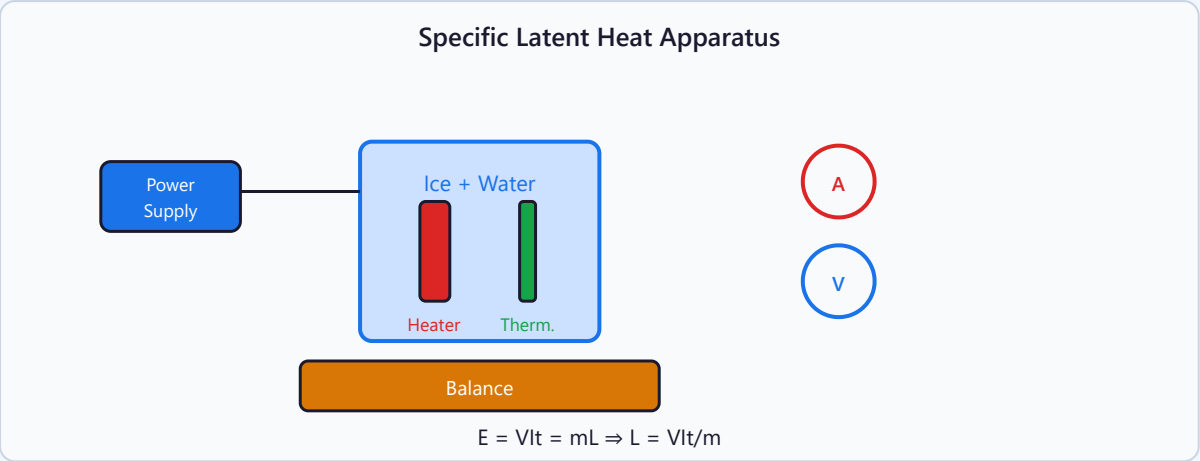
$$PV = nRT = \text{constant at fixed T}$$

P vs 1/V: straight line through origin. Gradient = nRT. Compare gradients at different temperatures.

Source of Error	Type	Improvement
Temperature not truly constant	Systematic	Use thermostatically controlled bath; allow equilibrium
Friction in piston	Systematic	Lubricate; tap gently to free
Gas leaks	Systematic	Check seals; use Vaseline
Reading V and P simultaneously	Random	Wait for stable readings before recording

Exam Tip: This practical is an extension of CP12 (Boyle's Law) with explicit temperature control. The gradient of P vs 1/V increases at higher temperatures because nRT is larger. This is consistent with the ideal gas equation.

CP16: Specific Latent Heat



Apparatus

- Immersion heater
- Beaker (insulated) with crushed ice or boiling water
- Thermometer
- Top-pan balance
- Ammeter, voltmeter
- Power supply, stopwatch
- Measuring cylinder (to collect meltwater or condensate)

Method (Latent Heat of Fusion)

1. Place crushed ice in a funnel over a beaker on a balance. Allow meltwater to drip.
2. First, measure the mass of ice that melts in a set time *without* the heater (control).
3. Repeat with the heater on. Record V, I, and time t.
4. The extra mass of ice melted due to the heater is $\Delta m = m_{heater} - m_{control}$.
5. Calculate $L_f = VIt / \Delta m$.

Method (Latent Heat of Vaporisation)

1. Boil water in an insulated beaker on a balance. Record initial mass.
2. Record V, I, and heat for a set time t.
3. Record final mass. Mass of water evaporated: $\Delta m = m_{initial} - m_{final}$.
4. Calculate $L_v = VIt / \Delta m$.

Safety

- Boiling water and steam can cause severe burns — handle with care
- Use insulated beaker; keep heater submerged
- Ensure electrical connections are dry
- Use a heat-resistant mat

Analysis

$$E = VIt = \Delta m \times L$$

$$L = VIt / \Delta m$$

Expected: L_f (ice) ≈ 334 kJ kg^{-1}; L_v (water) ≈ 2260 kJ kg^{-1}.

Source of Error	Type	Improvement
Heat loss to surroundings	Systematic	Insulate the beaker; use a control (no heater) for fusion
Splashing during boiling	Systematic	Use a gentle heat; cover beaker loosely
Heater not fully submerged	Systematic	Ensure heater stays immersed throughout
Mass measurement while steam escapes	Random	Read balance quickly; use a lid with a small hole

Exam Tip: The control experiment (melting without heater) is essential for latent heat of fusion. It accounts for heat absorbed from the surroundings. For vaporisation, energy is lost to heating the surroundings, so measured L_v is typically slightly higher than the accepted value.